Math 252: Midterm 1 **1 October, 2026**

Name: _____ **/ 100**

90 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 100 points possible.

1. Consider the region in the first quadrant bounded by the curves $y = \sqrt{1-x^2}$, $x = 0$, and $y = 0$.

   (a) (4 points) Sketch the graph of the region. Be sure to clearly label any important points.

   (b) (6 points) Write an integral that computes the volume of the solid obtained by rotating the region about the x-axis using the disk/washer method. **Do not compute the integral**.

   (c) (6 points) Write an integral that computes the volume of the solid obtained by rotating the region about the x-axis using the method of cylindrical shells. **Do not compute the integral**.

(d) (6 points) Compute one of the integrals from the previous two parts of this problem. You may choose which one to compute, but you must clearly indicate which one you have chosen.

(e) (4 points) How could you have calculated the volume from the previous parts **without using calculus**?

2. (6 points) Set up, but **do not evaluate** an integral that computes the arc length of the curve $y = e^{3x}$ between the points $(0,1)$ and $\left(\frac{1}{3}, e\right)$.

3. (10 points) A spring has a natural length of 2 meters. It requires 60 joules of work to stretch the spring to a length of 3 meters. How much would the spring stretch if you applied 15 joules of work? You may take Hooke's law ($F = kx$) as given, but must justify the rest of your answer with calculus.

4. Consider the region bounded by $y = x^2$ and $y = \sqrt{x}$.

   (a) (4 points) Sketch the region. Be sure to clearly label any important points.

   (b) (8 points) Set up expressions for the centroid (center of mass) of the region. **Do not evaluate** any integrals in your expressions.

   $\bar{x} =$ $\bar{y} =$

5. Evaluate the following integrals and antiderivatives.

   (a) (8 points) $\int_0^{\pi} x\sin(x)\,dx$

   (b) (8 points) $\int x^2 e^{3x}\,dx$

(c) (8 points) $\int \sin^2(x)\cos^2(x)\,dx$

(d) (6 points) $\int \sin^3(x)\cos^3(x)\,dx$

(e) (8 points) $\displaystyle\int \frac{dx}{x^2\sqrt{25-x^2}}$

(f) (8 points) $\displaystyle\int \frac{dx}{x^2-5x+6}$

6. BONUS (5 points) Evaluate $\int e^{5x} \sin(7x)\, dx$